S M Navin Nayer Anik

Associate Software Engineer II, AI/ML
B.Sc. in Computer Science & Engineering

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Career Summary

- Excellent team player with good problem-solving skills and self-motivation.
- Proactive and adaptable, able to thrive in a fast-paced environment.
- Machine Learning Engineer with strong problem-solving skills and a passion for innovation.
- Experienced in developing and optimizing machine learning models, working with large-scale datasets, and handling both structured and unstructured data.
- Proficient in deep learning frameworks like PyTorch and TensorFlow, with hands-on experience in model training, fine-tuning, and deployment.
- Well-versed in data preprocessing, feature engineering, and hyperparameter optimization.
- Skilled in time series forecasting, including building and deploying predictive models for real-world business applications.
- Experienced in using version control systems like Git, Bitbucket and working in collaborative environments with tools like GitHub.
- Passionate about writing clean, maintainable code and implementing scalable machine learning pipelines.
- Committed to continuous learning and staying updated with the latest advancements in AI and ML technologies.
- I am currently working as a junior software developer at Cefalo.

Tools & Technology

- Programming Languages: Python, Java, Assembly Language
- Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV
- Data Science & Analytics: Pandas, NumPy, Matplotlib, Seaborn
- Big Data & Cloud: Apache Spark, AWS S3, GCP, Google Colab, SageMaker, Spot Instances, Distributed Training, Lambda, EventBridge, CloudWatch, 3LC
- Databases: MySQL, MongoDB, SQL, Firebase, BigQuery
- Software Development: Git, Bitbucket, ROS, Shell Scripting
- Web Technologies: Node.js, Express.js, TypeScript, JavaScript, React.js, Redux Toolkit, Tailwind CSS, REST APIs
- Testing Library: Jest

Experience	*within Cefalo*
Project: *BioDrone* *Role: Associate Software Engineer II, AI/ML* *Duration:* *December 2025 - till now*	*Description:* *Worked with Biodrone, a Norway-based company specializing in advanced drone services and AI-driven aerial image analysis for forestry and agricultural applications.Contributing to the development and integration of machine learning models within the Biodrone Portal* *Tools / Technology: PyTorch, CNN, Transformers, Machine vision SageMaker, Docker, GitHub Actions, Distributed Training, AWS S3, EventBridge, 3LC*
Project: *AquaRobotics* *Role: Associate Software Engineer, AI/ML* *Duration:* *February, 2024 - July 2025*	*Description:* *Worked with Aqua Robotics, a Norway-based company providing robotic solutions for sea fish farming. Developed an image classification model for their underwater rover, addressing issues like overfitting and bias. Designed and implemented a GAN model to generate synthetic data, improving model performance. Integrated a new training method incorporating CI/CD pipelines using GitHub, Docker, and SageMaker Spot Instances, reducing training costs by up to 70%. Additionally, worked on distributed training to scale up training machines while minimizing cost and reducing training time.* *Tools / Technology: PyTorch, CNN, GANs, SageMaker, Docker, GitHub Actions, Distributed Training, Spot Instances, AWS S3, EventBridge, 3LC*
Project: *TicketCo Data* *Role: Associate Software Engineer II, AI/ML* *Duration:* *July, 2025 - September, 2025*	*Description:* *Worked with Tricketco Data on a proof-of-concept (PoC) project to develop a static rule engine designed to maximize ticket sales profit based on configurable business rules. Implemented a system leveraging NATS for real-time data streaming to publish and receive recommended ticket prices dynamically, adapting to market conditions and other influencing factors. The architecture was built with scalability in mind, enabling future integration of a dynamic AI-driven rule engine utilizing time series forecasting to generate price recommendations automatically. Additionally, designed the architectural blueprint for the future dynamic rule engine training pipeline, ensuring seamless transition from static to intelligent pricing models.* *Tools / Technology: fastAPI, NATS, Python, Microservices Architecture, Distributed Message Queues, Git*
Project: *Dobee* *Role: Associate Software Engineer II, AI/ML*	*Description:* *Worked with Dobee on a Generative AI project focused on building an intelligent and interactive story generation system. Designed an end-to-end pipeline to fetch raw data from BigQuery, use an AI model to generate insightful summaries and stories, and store the outputs in Firestore. Implemented a chatbot interface where users can provide additional context, allowing the system to regenerate more refined and context-aware*

Duration: *September, 2025 - December 2025*	*stories dynamically. This made the solution highly adaptive, enabling users to iteratively shape the generated content to match their specific needs. Additionally, leveraged Vertex AI for automated story generation and LangChain for context management and orchestration across the workflow.* *Tools / Technology: Vertex AI, LangChain, BigQuery, Firestore, TypeScript, Bitbucket*
Project: *Blogging Website (Backend and Frontend)* *Role:* Junior Software Developer *Duration:* September, 2023 – February, 2024	*Description:* *The website features JWT authentication for enhanced security. Users can create profiles, manage personal information, and securely update passwords. Registered users can create, update, and delete blogs, while all users can read and explore categorized blogs easily. It's a user-friendly platform for sharing thoughts and connecting with others, with the added assurance of JWT authentication. Unit tests are implemented for both frontend and backend components to ensure robust functionality and maintain code integrity throughout development and updates.* *Tools / Technology:* Express.js, Sequelize, React, TypeScript, Redux Toolkit, Tailwind CSS, React Router, Jest

Education & Certification

- B.Sc. in Computer Science & Engineering from BRAC University
- OWASP Top 10 for Large Language Model Applications Learning Path, awarded by SecureFlag
- AI for All: From Basics to GenAI Practice, awarded by NVIDIA

Hobby / Pet project

- Developed an Optimal Transport Theory-based GAN Network for Medical Image Augmentation.
- Developed a Real-time sleep detection and warning system, A real-time driver sleep detection system that alerts drivers if they feel sleepy while driving.
- Developed a new version of BRACU DUBURI
    - The first autonomous underwater vehicle in Bangladesh.
    - Capable of Underwater autonomous navigation.
    - A transfer-learning-based object detection algorithm is developed for the rover.
    - Can operate up to 30m underwater for 19 Minutes